FACULTY
OF EDUCATION
Masaryk University

Vim (Presentation #4)

Special Topics (CS3001)

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A Brief Introduction

They often say

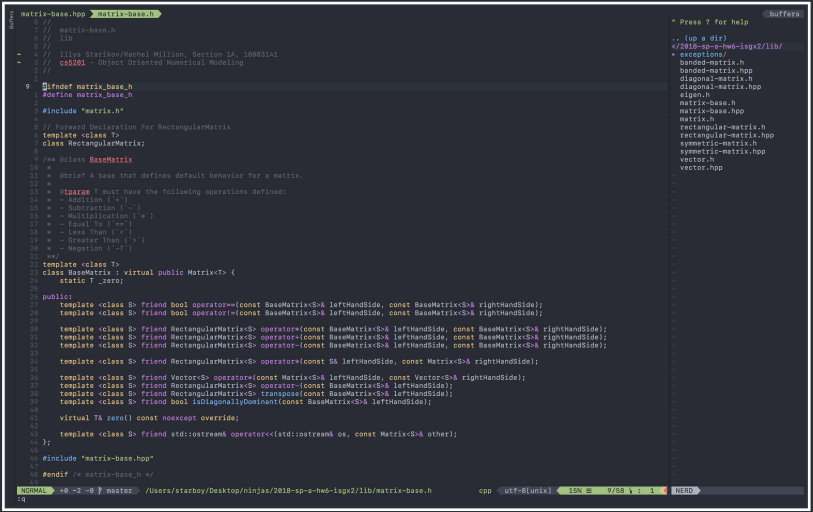
> *"A Developer Is Only As Good As His Tools"*

Although I don't agree with this sentiment, I always wanted to become better with tools I was constantly using. One of the tools I use most is my code editor: Vim.

A Brief Introduction

- Vim has been around since 1991, and is almost ubiquitous on all Unix machines. It has served mostly as a terminal text editor (see attached video).
- Also, Vim has greatly improved a lot since then, to incorporate many of the new features that IDEs and Text Editors have brought to the table.
- Along with that, a lot of projects have set out to make Vim look and feel more like a regular text editor (as apposed to a terminal editor).

A Brief Introduction




```cpp
//
//  matrix-base.h
//  lib
//
//  Illya Starikov/Rachel Million, Section 1A, 18083141
//  cs5201 - Object Oriented Numerical Modeling
//

#ifndef matrix_base_h
#define matrix_base_h

#include "matrix.h"

// Forward Declaration For RectangularMatrix
template <class T>
class RectangularMatrix;

/** @class BaseMatrix
 *
 *  @brief A base that defines default behavior for a matrix.
 *
 *  @tparam T must have the following operations defined:
 *  - Addition (`+`)
 *  - Subtraction (`-`)
 *  - Multiplication (`*`)
 *  - Equal To (`==`)
 *  - Less Than (`<`)
 *  - Greater Than (`>`)
 *  - Negation (`-T`)
 **/
template <class T>
class BaseMatrix : virtual public Matrix<T> {
    static T _zero;

public:
    template <class S> friend bool operator==(const BaseMatrix<S>& leftHandSide, const BaseMatrix<S>& rightHandSide);
    template <class S> friend bool operator!=(const BaseMatrix<S>& leftHandSide, const BaseMatrix<S>& rightHandSide);

    template <class S> friend RectangularMatrix<S> operator*(const BaseMatrix<S>& leftHandSide, const BaseMatrix<S>& rightHandSide);
    template <class S> friend RectangularMatrix<S> operator+(const BaseMatrix<S>& leftHandSide, const BaseMatrix<S>& rightHandSide);
    template <class S> friend RectangularMatrix<S> operator-(const BaseMatrix<S>& leftHandSide, const BaseMatrix<S>& rightHandSide);

    template <class S> friend RectangularMatrix<S> operator*(const S& leftHandSide, const Matrix<S>& rightHandSide);

    template <class S> friend Vector<S> operator*(const Matrix<S>& leftHandSide, const Vector<S>& rightHandSide);
    template <class S> friend RectangularMatrix<S> operator-(const BaseMatrix<S>& leftHandSide);
    template <class S> friend RectangularMatrix<S> transpose(const BaseMatrix<S>& leftHandSide);
    template <class S> friend bool isDiagonallyDominant(const BaseMatrix<S>& leftHandSide);

    virtual T& zero() const noexcept override;

    template <class S> friend std::ostream& operator<<(std::ostream& os, const Matrix<S>& other);
};

#include "matrix-base.hpp"

#endif /* matrix_base_h */
```

Prior Knowledge

- I’ve been using Vim since my freshmen year. However, I never worked with the advanced features of Vim, only the top level features.
- I knew that Vim had an expansive feature set, and I wanted to dive into that.

Motivation

- Upon graduation, I will likely be using Vim full time at my time at Garmin.

Goals

- To become more efficient at using Vim.
- To become familiar with the modern parts of Vim.
- To learn more of the keystrokes and shortcuts for Vim.

Resources

- A lot of the learning came from playing around with Vim. Anytime I found myself doing something repetitive, I simply tried to find a way to automate. Most of the time, there was a way.
- Book Resources (That I Actually Bought!):
  - Modern Vim
  - Practical Vim
- Online Resources:
  - The Vim Subreddit, for everyday improvements or general knowledge on Vim.
  - VimCasts, for when I want to get a deeper look into a subject.
  - Vim Adventures, a little game to help understand Vim through a text adventure.

Goal Accomplishment

- I am much more fluent in using Vim, using shortcuts that significantly improve my workflow.
  - See attached videos for details.
- I have also learned to use plugins, to do completions.
- I have also learned to use many of the different parts of Vim I was unfamiliar with, such as:
  - Buffers
  - Registers
  - Advanced Search
  - Tags

  And a lot more!

Side Note

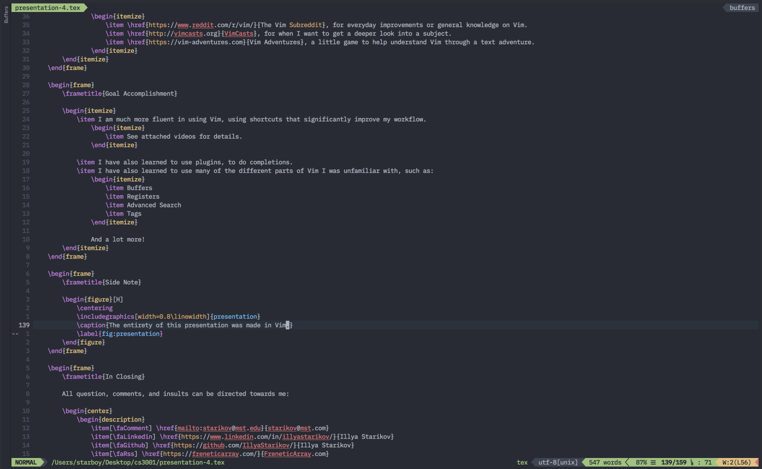


Figure: The entirety of this presentation was made in Vim.

In Closing

All question, comments, and insults can be directed towards me:

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